PORTALDOT HACKER HOUSE 2026

SEAM

The execution layer for Portaldot.

One Manifest in. One receipt out. The wallet holds the keys.

TRACK
Modular Cloud & Middleware

TEAM
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LIVE
seam-portaldot.vercel.app

REPO
github.com/rishu4436/seam-portaldot

Apache-2.0 · Hosted V3 testnet only · Mainnet sends are blocked

THE PROBLEM

A Portaldot intent is not one call.

01

Several surfaces, one intent

Assets, NFTs, Revive and batch_all each encode differently. A runtime change makes yesterday's hand-built bytes wrong.

02

The signature can drift

The user approves one plan. The signed extrinsic can name a different recipient, amount, asset or nonce.

03

Finalized is not proven

A transaction in a block is not proof the intended effect happened. Missing evidence has to stay unknown.

WHY PORTALDOT

The product only exists because of this runtime.

Portaldot V3 puts pallet calls and Revive on one chain. SEAM reads that chain's live metadata and compiles the exact calls. Indexes are never hard-coded. If the metadata has no call, SEAM refuses to invent the layout.

- **Assets**
  Create, mint, transfer
- **NFTs**
  Create and mint
- **Revive**
  EVM deploy and call
- **batch_all**
  Atomic only after rollback proof

SCOPE

Every send in this deck is the hosted Portaldot V3 testnet, spec 3002, genesis 0xd4f9…83bd. The console refuses mainnet and any other chain before the wallet opens, and again before submit. Mainnet was read at block 2,721,455. Nothing was submitted there.

HOW IT WORKS

Compile, bind, sign, prove.

1

Manifest

What should happen, as data.

2

Compile

Exact calls from live metadata.

3

Policy

Caps, allow-lists, runtime lock.

4

Wallet

The user signs. SEAM holds no keys.

5

Gate

Bytes, nonce and dry run, again.

6

Receipt

Finalized effect, or unknown.

Two plan modes

Sequential sends one transaction per step. Atomic uses Utility.batch_all only as atomic_verified, and only after rollback has been shown on that runtime. An unproven effect stays unknown. It is never marked success.

WHAT IT REFUSES

Failed pre-submit checks stop the broadcast

POLICY

- Amount, fee and weight caps
- Recipient and asset allow-lists
- Contract and selector allow-lists
- Pinned ABI and message name

SUBMIT GATE

- approval_reused / approval_expired
- runtime_changed
- signed_bytes_mismatch
- signed_dry_run_failed

An unproven effect stays unknown. A labelled test-only bypass produced the recorded batch rollback.

PROOF IT WORKS

Hosted Portaldot V3, not a local mock.

18 finalized txs on 19 Oct

3002 portaldot-v3 spec

0 mainnet transactions

Sequential plan	#102653 pay #102657 mint	Receipt re-checked
atomic_verified	#102671 Utility.batch_all	Rollback shown on this runtime
Rollback probe	#102665 whole batch failed	Balances unchanged; fee only
Revive call	#102853 ContractEmitted	Effect proven only with that log
Wallet transfers	#118310 #121390 / #121394	polkadot.js extension
Native swaps	#341514 tPOT to DUCAT #341539 back	6 Nov, canonical execution

HONEST BOUNDARY

Shown on V3, and not shown.

SHOWN

- Metadata compile, policy, approval, submit gate
- Assets, NFTs, sequential and atomic_verified plans
- EVM deploy, and a revive.call that emitted a log
- Extension-signed asset transfers
- Native swaps re-read from finalized blocks

NOT SHOWN

- No mainnet submission
- No ink! or PolkaVM path on V3
- No cross-VM atomicity
- No fresh account mapping
- A revive.call with no log stays unknown
- Contract deploy was setup, outside policy

ECOSYSTEM

Other Portaldot apps can integrate the SDK

@seam/compiler

Manifest to exact calls

@seam/policy

Rules, approval, gate

@seam/chain

Discover, sign, submit

@seam/sdk

Receipts and explanations

@seam/console

Reference workspace

seam CLI

compile, check, verify

Apache-2.0. The consumer example uses package APIs from locally packed archives.
The live workspace runs on the hosted V3 testnet.

CONTINUED SUPPORT

Worth carrying past demo day.

Who it is for

Teams shipping Portaldot apps, and agents that must execute a constrained workflow and hand back a receipt a third party can re-check.

What stays free

The compiler, policy, SDK and console stay open source. SEAM does not custody keys and does not add a token.

Commercial validation

Hosted monitoring and receipt checks already run on testnet. Customer demand and pricing remain unvalidated.

Next milestones

Expand wallet coverage. Pilot the SDK in one external app. Review DeFi entry and full exit paths. Enable mainnet only after network-specific rehearsal and independent security review.

WHERE TO LOOK

SEAM

One approved intent becomes a receipt the chain can still explain.

NATIVE SWAPS

Open the live native swap workspace

TRANSACTION BUILDER

Open the live transaction builder

REPOSITORY

https://github.com/rishu4436/seam-portaldot

NATIVE SWAP EVIDENCE

Finalized swaps: blocks 341514 and 341539

DEMO VIDEO

Watch the product walkthrough (2:30)

LICENSE

Apache-2.0 · Rishu Kumar Gupta